



THE MAIN TITLE OF THE THESIS

THE SUBTITLE OF THE THESIS IF IT HAS ONE

YOUR NAME

THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE

DEPARTMENT OF
COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE
SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
TILBURG UNIVERSITY

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DATE

May 7, 2026

WORD COUNT

your word count should be here, ex 12000

ACKNOWLEDGMENTS

Some room for acknowledgements.

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Abstract

This is where the abstract goes. Don't forget to change the variables in `main.tex` to change all general placeholders shown in this document. The `frontmatter.tex` file should be left alone. The abstract should summarize in 150-250 words the contents of the thesis, mentioning: 1) the research goal, 2) the methods employed, 3) data collected or dataset/s used, 4) the main findings.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Checklist for creating a Data Source, Ethics, Code, and Technology (DSECT) statement: Please provide answers to all below questions for transparency purposes when writing your ethical statement. There is no one-fits-all solution to ethical statements, and they heavily depend on the nature of the study. After the questions below, an example DSECT statement is provided.

- Data Source
 - o Who is the owner of the data?
 - o Does the thesis project involve collecting data from human participants or animals?
 - o Does the owner of the data give consent?
 - o How and through what channel is the data acquired?
- Figures
 - o Did you create all the images and figures?
 - o Do you have consent for the images that you did not generate?
- Code
 - o Did you use parts of the code from another study/someone else?

- o Did you list, including version number, all libraries and frameworks used?
- Technology
 - o Did you use any tools or services to paraphrase the given text (for example a thesaurus or the Academic Phrasebank)? Please name them.
 - o Did you use any tools or services to check spelling or grammar? Please name them.
 - o Did you use any tools or service to typeset the given text? Please name them.
 - o Did you use any reference management software other than the latex template? If so, please name it.
 - o The following **statement should be included**: During the preparation of this work, the author used [NAME TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the thesis.

A DSECT Statement can be generated by answering the questions listed above and excluding the questions when not applicable.

1.1 Source/Code/Ethics/Technology Statement Example

Data Source: The (DATA) has been acquired from the (DATA SOURCE) through an online request. The obtained data is anonymised. Work on this thesis did not involve collecting data from human participants or animals. The original owner of the data and code used in this thesis retains ownership of the data and code during and after the completion of this thesis. However, the institution was informed about the use of this data for this thesis and potential research publications. All the figures but Figure A and B belong to the author. To use Figure A and Figure B, the author received written permission via an email from FIGURE OWNER (AFFILIATION). The thesis code can be accessed through the GitHub repository following the link [CODE LINK]. Part of the CODE has been adapted by the author from (SOURCE, licensed under a CCo BY 4.0). The reused/adapted code fragments are clearly indicated in the notebook. In terms of writing, the author used assistance with the language of the paper. A generative language model (MODEL NAME + SOURCE) was used to improve the author's original content, for paraphrasing, spell checking and grammar. No other typesetting tools or services were used.

2 INTRODUCTION

Here you write the introduction. Start with briefly describing the research domain, why the topic is relevant, and current state-of-the-art by referencing pertinent research articles. Then, state your specific research goals/questions (you can separate them as shown below), followed by a one-paragraph summary of your findings. Discuss how answering your research questions can produce scientific relevance (domain insights) and societal relevance (real-world applications and/or benefits).

You can refer to chapters like the current Section 2. If you want to list research questions it's probably best to use quote and/or itemize, like so:

To what extent we define a main research question?

The sub-questions can be listed separately, as such:

RQ1 *How does sub question one influence the main RQ?*

RQ2 *To what degree does sub question add anything?*

Or, format it as you desire (tip: you can nest itemize as well). You can alternate *emph* and **textbf** however you wish. This should cover most of the things required for the introduction.

3 RELATED WORK

Here you provide a synthesis of the existing literature, comparing and drawing parallels across studies to establish the state-of-the-art. Cite relevant theories, models, and recent work. Describe what prior research has found, any contradicting findings or competing approaches. Identify any gaps or limitations in existing research and how your thesis potentially addresses these gaps in terms of methodology, data used, etc.

Copy paste BibTeX code¹ and put it in references.bib. After, you can cite some work (MacKay & Mac Kay, 2003) – using \citep. You can refer to the author of e.g. Minsky (1961) directly like using \cite (this does not work when using bracket-citation). If you use bracket-style,² you might want use \citeauthor when citing, like: see Ananny and Crawford Ananny and Crawford (2018). If you want to add pages you can use brackets in \citep[[p. 5]{mackay2003information}, which looks like: (MacKay & Mac Kay, 2003, p. 5). The first brackets can be used for things like *see*, and *e.g.*. If you want to cite multiple authors, simply comma-separate them (\citep{minsky1961steps,mackay2003information}) and

¹ Using e.g. the quote icon in GScholar, then BibTeX at the bottom.

² Find the natbib part in the main.tex L^AT_EX script.

it will aggregate them automatically (MacKay & Mac Kay, 2003; Minsky, 1961).

4 METHODS

4.1 Dataset

The CogPilot dataset(Rao,2022) was used in this project and are publicly available via PhysioNet(Golbeger, 2000). The

For machine learning studies, the sub-sections may include: **Dataset/s** (i.e., source, size, features, and descriptive statistics where appropriate), **Feature Extraction** (if applicable), **Machine Learning Setup** (i.e., dataset splits, cross-validation approach, preprocessing steps, algorithms used, hyperparameters tuned), **Evaluation Metrics** (i.e., how model performance was assessed), **Implementation** (i.e., programming languages, packages used).

If you define any equations ($\begin{equation} \dots \end{equation}$), you probably might want to define everything using math operators (e.g., $\$D\$$) and cite the work (!). So for example, following Lewis and Gale (1994), representing a document $d \in D$ as $\text{tf}(d)$, we define a probabilistic model $(d|Y = y)$ for all documents in class y , and select y most likely to generate d :

$$\hat{y} = \underset{y}{\operatorname{argmax}} P(d|y) \cdot P(y) \quad (1)$$

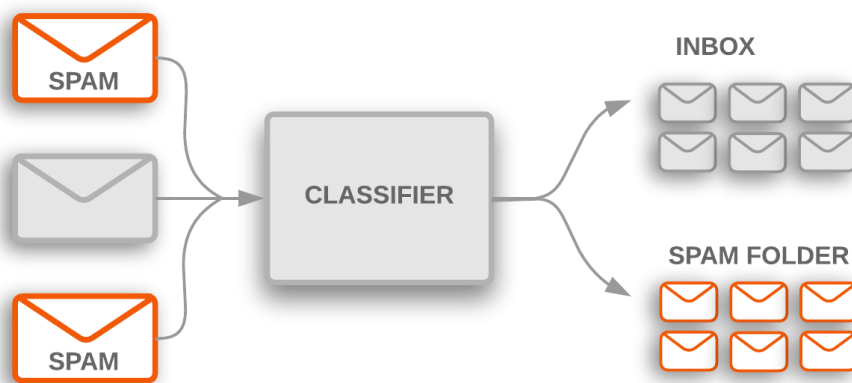


Figure 1: Spam classification example. Source: Google (CC BY 4.0).

Table 1: Best scoring models classifying bots on Twitter and Facebook, respectively. F_1 scores report positive (bot) class. Outline text left (l) and numbers right (r).

PCA	Models	F_1 score	
		Twitter	Facebook
300	Linear SVM ($C = 0.1$)	0.51	0.91
	Random Forest ($S = 5, F = 5$)	0.71	0.85
	Naive Bayes	0.61	0.73
500	Linear SVM ($C = 0.1$)	0.55	0.84
	Random Forest ($S = 5, F = 5$)	0.76	0.71
	Naive Bayes	0.41	0.64
	Majority	0.50	0.60

With this we can detect spam (see Figure 1) or bots. Note that the figures (and tables for that matter) might not always be placed in this section. $\LaTeX$ typically determines where to best put your objects. However, the [H] option from the float package forces figures to appear at this exact location rather than floating to other sections. **NOTE:** this Figure has a Creative Commons license; you cannot re-use other authors' figures without explicit permission or permissive licensing (as this would mean copyright infringement). You can refer to the equations as well (Equation 1).

5 RESULTS

Report your main findings here (e.g., model performance metrics, statistical test outcomes) and highlight important or interesting patterns. All information in tables and figures should be summarized/interpreted in text focusing on notable trends and comparisons. Where appropriate, conduct further exploration through statistical analysis, confusion matrices, error analysis, or stratified results (e.g., performance across sub-groups, classes, or conditions) to reveal patterns not obvious from overall results.

Figures and tables: Use high-quality visualizations with labeled axes, clear legends, and informative captions (see Figure 2). For table formatting, see Table 1 as an example. Highlight important scores with `\textbf{}`, use `booktabs` commands for structure: `\toprule \midrule \bottomrule`. APA does not allow vertical lines.

5.1 Some Model

If you have anything specific to talk about, use subsections, and refer to them as Section 5.1. Don't use paragraphs or subsubsections.

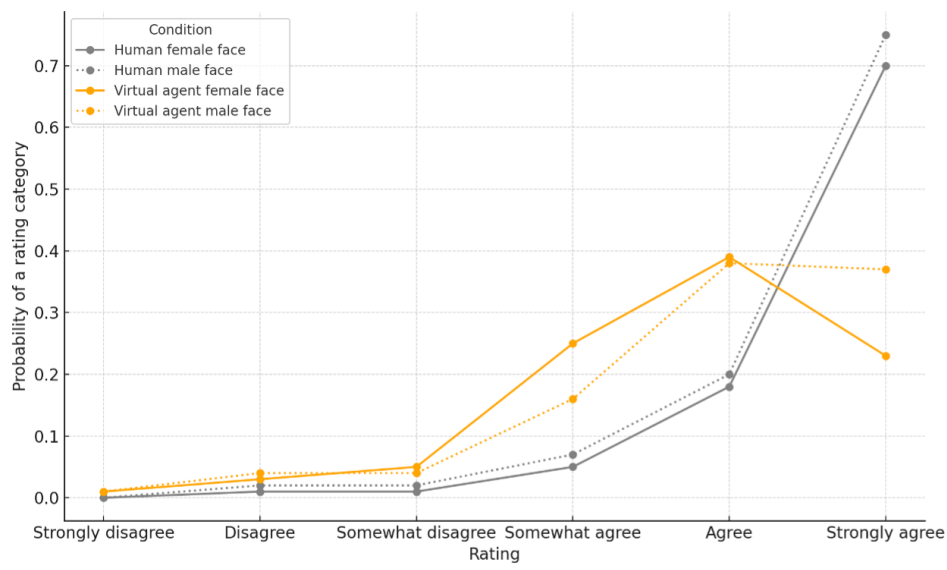


Figure 2: Cumulative probabilities based on Image Type (human and virtual agent) and Gender (female and male) for each outcome of a rating: from “Strongly disagree” to “Strongly agree”.

6 DISCUSSION

Here you interpret your findings in light of your research goals and the broader literature. Begin by briefly restating your research goal and summarizing your main findings. Then, discuss how these results relate to the existing research context established in your Introduction, i.e., do they confirm, contradict, or extend prior work? What is the specific contribution of your thesis?

Discuss your findings in the same order presented in the Results section. Offer interpretations for surprising or unexpected outcomes relying on the existing literature. Acknowledge any limitations of your methods, models, or data that may have affected your results, and explain how these limitations impact the validity and scope of your findings. Where relevant, suggest alternative approaches that could address these limitations.

Conclude by identifying directions for future research based on your findings, what questions remain unanswered? What new questions has your thesis raised?

7 CONCLUSION

Here you provide a concise summary of the overall work in your thesis. Essentially, you want to answer: what are the key takeaways? Clearly

articulate the implications of your work: what does it mean for the field, for practice, or for theory?

Keep this section focused and avoid introducing new information, as that belongs in the Discussion.

REFERENCES

- Ananny, M., & Crawford, K. (2018). Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. *New Media & Society*, 20(3), 973–989.
- Lewis, D. D., & Gale, W. A. (1994). A sequential algorithm for training text classifiers. In *Sigir'94* (pp. 3–12).
- MacKay, D. J., & Mac Kay, D. J. (2003). *Information theory, inference and learning algorithms*. Cambridge university press.
- Minsky, M. (1961). Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1), 8–30.

APPENDIX A

If you have nothing to append: remove this. You can do a page referral for these, like: Appendix A (page 7).

APPENDIX B

And this!